

A Millimeter-Wave CMOS Wideband Concurrent Dual-Band Dual-Beam Phased array Receiver Front-End

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Summary/Abstract

This document presents the design, implementation, and characterization of a millimeter-wave CMOS wideband concurrent dual-band dual-beam phased array receiver front-end targeting advanced 5G wireless communication systems. The receiver front-end is fully integrated using Global Foundry 22 nm FDSOI CMOS technology and is developed to facilitate multi-band MIMO communication in multi-antenna systems, allowing for faster and more reliable connectivity and supporting emerging applications such as smart cities and autonomous vehicles.

The architecture is designed for concurrent operation across dual bands (24–28 GHz and 37–40 GHz), offering dual output streams and supporting both carrier aggregation and multi-beam functionalities. The system comprises four input channels and employs a fully-connected RF beamforming structure, which allows for spatial selectivity, null tuning, and efficient spectrum interference cancellation right after the low-noise amplifier (LNA) stage. Key circuit blocks include four LNAs, four five-bit active phase shifters with gain-compensated RC poly-phase filters for phase/gain accuracy, quadrature networks, eight vector modulators, and two wideband distributed active combiners. A unique aspect of the design is the phase shifter enabling simultaneous RF beamforming for two independent output streams, and the use of a novel three-winding transformer load allowing the LNA and receiver to operate in reconfigurable modes with high image rejection.

System-level simulations demonstrate effective single-beam and dual-beam steering, robust link budget analysis, and phase accuracy impact on error vector magnitude (EVM). Circuit-level implementation details highlight techniques for reducing noise figure, optimizing layout for parasitic reduction, and maintaining phase and amplitude accuracy. Fabrication and measurement results include characterization setups for standalone LNAs and phase shifters, over-the-air RF measurements, and beam steering capabilities.

In summary, this work proposes and validates a novel, highly integrated mm-wave receiver architecture capable of concurrent dual-band, dual-beam operation with innovative circuit solutions for wideband RF beamforming. The approach is verified through extensive simulations and practical measurements, showcasing its potential for future 5G multi-antenna communication systems.

Key References

1. R. Lu, C. Weston, D. Weyer, F. Buhler, D. Lambalot, and M. P. Flynn, “A 16-element fully integrated 28-GHz digital RX beamforming receiver,” *IEEE Journal of Solid-State Circuits*, vol. 56, no. 5, pp. 1374–1386, 2021